



Document Effective Date  
10/12/2023

IS HAZARD & OPERABILITY (HAZOP) REVIEW GUIDELINE

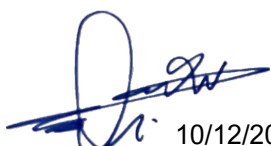
DOC NO: IS-HSE-GDL-318

Rev: 01

Quality Reviewed By: OIB/4

 Digitally signed by  
Moklis Miah  
Date: 2023.12.07  
11:49:57 +03'00'

| CUSTODIAN DEPARTMENT | DOCUMENT AUTHOR                                                                                                   | DOCUMENT OWNER                                                                                                    |
|----------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Add Department Name  | <div>Digitally signed by<br/>Tigor Maruli<br/>Date: 2023.12.07<br/>13:21:44 +03'00'</div> <div>Tigor Maruli</div> | <div>Digitally signed by<br/>Tigor Maruli<br/>Date: 2023.12.07<br/>13:21:57 +03'00'</div> <div>Tigor Maruli</div> |
|                      | OIH/43                                                                                                            | A/OIH/4                                                                                                           |

| FINAL APPROVAL                                                                                                |
|---------------------------------------------------------------------------------------------------------------|
| <div><br/>10/12/2023</div> |
| OIH                                                                                                           |

|                                                                                   |                                                                                                                            |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|  | <div><div>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</div><div>DOC NO: IS-HSE-GDL-318</div><div>REV. 01</div></div> |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|

TABLE OF CONTENTS

Page No.

1.0 PURPOSE ..... 3

2.0 SCOPE AND APPLICABILITY..... 3

3.0 GUIDELINE ..... 3

    3.1 GENERAL ..... 3

    3.2 HAZOP TEAM..... 4

    3.3 METHODOLOGY ..... 5

4.0 REFERENCES TO OTHER DOCUMENTS ..... 17

5.0 DEFINITIONS ..... 17

APPENDIX..... 19

    APPENDIX A - PARAMETERS & GUIDE WORDS ..... 19

    APPENDIX B - DEVIATIONS & POTENTIAL CAUSES ..... 20

    APPENDIX C - OTHER CONSIDERATION FACTORS ..... 21

    APPENDIX D - CONDITIOINING FACTOR TYPES ..... 21

    APPENDIX E - SAMPLE HAZOP WORKSHEETS ..... 24

    APPENDIX F - REVISION HISTORY LOG ..... 26

    APPENDIX G - LIST OF ABBREVIATIONS..... 26

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

## 1.0 PURPOSE

This Guideline is intended to provide guidance on the Qatar Energy/ Idd El Shargi Procedure for Process Hazard review, required in IS-HSE-PRC-308.

The General objectives of the QatarEnergy/ Idd El Shargi Risk Management Guidelines within the QatarEnergy/ Idd El Shargi Risk Management process are to:

- Allow a clearer definition of requirements with regard to the QatarEnergy/ Idd El Shargi RM program and related Policies and Procedures;
- Allow specific topics to be addressed in detail relevant to the application of the RM element, including examples where relevant, and
- Act as a basis for further information, understanding and training.

Note: These Documents should be considered as Guidelines only and not mandatory unless otherwise stated, the use of the words, 'shall' and 'should' will indicate a mandatory requirement. For further clarification on content and intent contact the QatarEnergy/ Idd El Shargi Technical HSE Engineer.

The purpose of this Guideline is to provide guidance on when and how to apply CMEs when calculating Consequences associated with an Event Scenario; and to ensure that, when evaluating Risk, a consistent approach to the application of Critical Mitigation Elements is applied.

## 2.0 SCOPE AND APPLICABILITY

This Guideline applies to all personnel within QatarEnergy/ Idd El Shargi who are involved in Risk Management, and applies to Employees, contractors, and subcontractors, whether engaged directly by QatarEnergy or by others, who will perform work or services of any kind at any QatarEnergy/ Idd El Shargi facilities and/or locations. The Guideline is applicable to all QatarEnergy/ Idd El Shargi facilities; process and utility systems. Coverage will include all systems containing hydrocarbons or other hazardous fluids which may pose a risk to personnel, the environment, capital as well as utility systems e.g. vent/flare system, open & closed drainage systems, fire protection systems, emergency control system.

## 3.0 GUIDELINE

### 3.1 GENERAL

The technique for a Hazard & Operability Review, (HAZOP), has been developed to provide a formal, structured, systematic, team based, method for identifying the causes and analysing effects, of deviations from the design intent or from safe plant operations associated with an existing or new system, facility or equipment and to identify the necessary control measures to safely manage any undesired consequential effects.

HAZOP reviews are basically a communication exercise. Information is presented, discussed, analysed and recorded. Specifically the safety aspects are identified, to determine if adequate design measures have been taken to prevent major accidents. Communication and evaluation are the prime facets of the Procedures. HAZOP reviews follow a definitive guideword approach, step by step with the objective to provide assurance that catastrophic incidents will be avoided during the lifetime of the facility from the processes under review.

|                                                                                   |                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b> <span style="float: right;"><b>REV. 01</b></span> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

A HAZOP is a living Document for a facility. As changes are made to a facility or its Procedures the HAZOP will be updated to represent the current facility. Process hazard analysis reviews are also required to be updated and revalidated every five years as a minimum, by QatarEnergy/ Idd El Shargi.

A core HAZOP is undertaken to verify the design and should be carried out during the Detailed Design phase of a project. Revalidation HAZOPs should be undertaken routinely, with a maximum periodicity of 5 years, to ensure that the critical mitigation measures applied at the previous review still apply and that no new hazard and operability issues have been introduced during the intervening period. It is always to be remembered that a HAZOP should be considered a design verification rather than an opportunity for 're-design'.

HAZOP studies may also be undertaken to verify the design for complex plant changes (QTMs) the MOC committee identify the requirement for those studies.

Coarse HAZOPs may be undertaken during the Preliminary Design phase of a project, however, due to low level of project definition at that project stage, the Coarse HAZOP may take the form of a What-if Review. For further information on What-if Studies reference should be made to The RM What-If Guideline, Appendix B (QP-ISND-RMG-HSE-308-50).

## **3.2 HAZOP TEAM**

### **3.2.1 Team Structure**

All core HAZOPS should be led by an 'independent', PHR Facilitator, endorsed by the QatarEnergy/ Idd El Shargi Technical HSE. The PHR Facilitator will be assisted by an independent Scribe.

#### **Core and Non-Core Teams**

HAZOP Team membership will be categorised into Core and Non-Core Teams. Non-core team members may be requested to participate on an 'as required' basis.

The core team shall consist of:

- Project Representative
- Senior Process Engineer
- Instrumentation/Control Engineer
- HSE Representative
- Operations Representative
- Maintenance Representative

**Non-core team** members who may be called upon as required include:

- Asset Integrity Representative
- Specialist Engineering Representatives (Technical Authorities)
- Consultant/Third Party Specialists
- Vendors Representatives

Where the HAZOP Review is on a contracted engineering project the core team will incorporate additional representation from the Contractor as follows:

- Project Engineer

|                                                                                   |                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b> <span style="float: right;"><b>REV. 01</b></span> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

- Process Engineer
- Instrument Engineer (optional)
- Loss Prevention Engineer

Non-core team will consist of:

- Other specialist Engineers as and when required by HAZOP.

It is essential that the team composition be appropriate for the plant or design being studied.

All persons, attending a HAZOP Review should be made available for its full duration, without substitution. Continuity of the team is critical to a consistent approach and understanding of the previous HAZOP agreements and discussions.

All HAZOP team members should ideally be formally trained in the HAZOP process by means of an internationally recognized HAZOP training course.

### **3.2.2 PHR Facilitator**

The PHR Facilitator should be an experienced engineer with relevant knowledge in the oil and gas industry, but a generalist rather than an 'expert', he should be experienced, qualified and certified in Facilitating HAZOP studies of a similar level of complexity.

Responsibility for the conduct of the HAZOP lies with the PHR Facilitator. The PHR Facilitator has the responsibility to ensure that the HAZOP proceeds in an effective and efficient manner, maintaining the essential flow of the Review, its robustness and in ensuring the active involvement and participation of all members of the HAZOP team.

The PHR Facilitator, with the assistance of the scribe, is responsible for preparation of the HAZOP report and the HAZOP Record sheets, for agreement by and acceptance of the team.

The PHR Facilitator does not have responsibility for implementation or close out of action items arising from the HAZOP.

## **3.3 METHODOLOGY**

### **3.3.1 Review Definition**

A HAZOP Review is a team based activity conducted in line with a Procedure for the systematic and critical examination of a system, subsystem or facility design, to identify potential hazards that could have an impact on each specific consequence category:

- Personnel;
- Environment;
- Property (or Facility) Damage & Loss of Revenue (PD/LOR);
- Impacts Outside a facility;

As well as:

- Operability of the facility or:
- Maintainability of the facility.

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

and to enable identification of corrective actions in the form of additional controls and recommendations to be put in place which will either mitigate or eliminate such hazards.

As with all PHAs, a HAZOP should have a clearly defined statement of requirements defining the purpose of the Review, the limits of its scope and its specific objective. Such a statement is required to ensure that the Review focus is clearly understood and that the review team keep to that focus. The PHR Facilitator may use the Review definition during the HAZOP review to assist in preventing the team from becoming diverted from the true focus of the Review.

### **3.3.1.1 Purpose**

The purpose should identify the reason why the HAZOP Review is being performed, for example, to meet the requirements of company Standards for risk evaluation and risk management. It affects the way a Review is performed, for example, the types of hazards to be included and the types of consequences to be addressed. Its definition helps to ensure the Review outcome is consistent with the intention for the Review.

### **3.3.1.2 Scope**

The scope of the HAZOP should identify the limits of the Review, clearly defining, what is to be included within the scope of the Review and what the boundary limits for the review are, the design intent, the level of detail, the level of causality, the assumptions and the exclusions to be applied.

### **3.3.1.3 Objective**

The objective of the HAZOP Review should be an expansion of the HAZOP purpose. The initial establishment of purpose, scope, and objectives, is very important and should be precisely set down so that it will be clear now and in the future as to what was and what was not, included in the Review.

## **3.3.2 HAZOP Preparation**

A significant amount of preparatory work is required both by the HAZOP team and by the discipline Engineers concerned, prior to each session. This work typically comprises;

- Collecting the relevant data,
- Validation of information
- Converting the data into a suitable form,
- Node preparation
- Nodding P&IDs
- Planning the sequence of the review,
- Arranging and scheduling the necessary meetings,
- Appropriately pre-populating as much of the HAZOP template as possible with available basic data which does not require the HAZOP team's input but saves valuable time during the review process e.g. purpose, objectives, scope of the PHR Review; interfaces; key Documents/drawings; node data etc.
- Issuing the data pack to team members,
- Briefing of team members.

### **3.3.3 Data Collation**

|                                                                                   |                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b> <span style="float: right;"><b>REV. 01</b></span> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

The Process TA will be responsible for assembling the relevant data to meet the requirements of the PHR Facilitator. Individual discipline Lead Engineers will be responsible for supplying the data to the Process TA and where necessary, advising on the available data and other issues of interest. P&IDs and other documentation used for HAZOP must be accurate, up-to-date and marked as 'issued for HAZOP'.

A HAZOP dossier containing the above listed Documents should be issued to all participants for their review and preparation for the HAZOP at least 1 week prior to the start of the HAZOP Review.

The following Documents are required by the HAZOP team:

- PFDs,
- P&IDs for Process & Utilities,
- QatarEnergy/ Idd El Shargi HSE Risk Matrix and Response Chart
- Heat and Mass balances,
- Plot Plans / General Equipment Arrangements & Elevation Drawings,
- Material Specifications and Selection Charts,
- Cause and Effect Diagrams,
- Process Design Basis,
- Process Operation & Control Philosophy,
- Operations and Maintenance Manual,
- Material Safety Data Sheets.

The following Documents may, if available, also be consulted by the HAZOP team. Availability will depend upon the developmental stage of design of the facility/plant being studied.

- Safety philosophy,
- SIL Review report,
- Process Safeguarding Flow Diagram,
- Electrical Single Line Diagrams,
- Equipment Specifications,
- Vendor Data and Drawings,
- Fire and Gas Detection Drawings,
- Hazardous Area Classification Drawings.

Additional Documents may be called for during the HAZOP to amplify or verify information requested by the HAZOP team.

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

### 3.3.4 Data Presentation

To save time during the Review meetings, some Documents used during the review may be marked up with additional information, e.g. P&IDs annotated with normal and design conditions, instrument set points etc.

Any marking should be done by the relevant discipline engineer either on his own behalf or at the request of the Independent PHR Facilitator.

### 3.3.5 Phasing the Activities

The PHR Facilitator will plan the sequence of the Review, usually the facility or plant will be separated into systems and subsystems. The sub-systems will be separated into defined nodes on the relevant P&IDs which can then be examined in a logical fashion by the HAZOP Team, so that the results of earlier studies contribute to subsequent studies, this work should be prepared in advance of the Review.

### 3.3.6 Scheduling the Meetings

The PHR Facilitator will be responsible for arranging the Review sessions and any other relevant meetings. The PHR Administrator will be responsible for booking the venue, alerting necessary personnel to their attendance requirements, arranging for the necessary visual aids, black/(white) boards or flip pads, and computer/IT facilities etc to be made available in due time prior to the start of the Review.

It is an essential requirement that the HAZOP Review should maintain continuity. Accordingly, the sessions should follow on, on a daily basis. The HAZOP process is intensive and the quality of the Review sessions and effectiveness of the team will be negatively impacted by Review durations exceeding more than approximately 6 Review hours per day. Short breaks for refreshment and a lunch break are essential to the continued efficiency and effectiveness of the Review especially when undertaking core HAZOP studies on major facilities which due to their scope require to be conducted over a long period of Review sessions.

### 3.3.7 Review Sessions

#### 3.3.7.1 General

A basic requirement of the HAZOP is that the team question every aspect of the process and utilities design to determine how deviations from the design intent and/or from normal operations may lead to hazardous situations and/or operability problems.

At the start of the HAZOP the PHR Facilitator should provide the team with a briefing on the HAZOP process and the requirements and responsibilities of HAZOP team members and the general etiquette expected during the review sessions. e.g. Refraining from the use of mobile phones and/or other personal electronic devices for Non HAZOP related matters. Holding side discussions without going through the PHR Facilitator.

The PHR Facilitator will help the Team in establishing agreed start and break times and the Team members should ensure that they are available to start each session and reconvene after breaks at the agreed upon times.

The PHR Facilitator should clarify that problem-solving or re-designing during the sessions is discouraged however, simple problems may well be solved and decisions taken to recommend modification to the design.

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

The PHR Facilitator has responsibility to plan and control the sequencing of the Review. The strategy involves the process engineer providing the team with a brief description of the system, after which the PHR Facilitator will select the first item of equipment or process path (Node) to be examined.

### 3.3.7.2 Nodes

The division of the facility/design modifications under HAZOP review, should be defined into appropriately sized segments termed as; NODES. A node may contain process lines and/or equipment for HAZOP Review.

The function/design intent of the equipment or system within the Node being studied should be clearly and concisely explained and as a minimum address: Material, Activity, Source, Destination, Key characteristics. Its explicit purpose agreed by the team.

### 3.3.7.3 Conditioning Factors<sup>1</sup>

For Brownfield and Revalidation HAZOPs, any conditions affecting equipment and piping (i.e. General condition of equipment / piping, Deferrals, Documented Failures / Problems, Temporary repairs / wraps, NROs (non-routine operations), LTIs (long term inhibits), Outstanding actions from related PHAs, PSSRs and Punch lists), which may impair the robustness of an Intrinsic Control should be identified as a Conditioning Factor and the related CME disregarded as an appropriate control or reduced in status to a KME depending on the severity of the conditioning factor<sup>2</sup>.

### 3.3.7.4 Parameters & Guide Words<sup>3</sup>

The usual parameters applied to all Nodes are Flow, Pressure and Temperature. Additional Parameters include; Level, Viscosity, Composition, Start-up /Shutdown, etc. may be applied as required by the contents of each Node.

The PHR Facilitator should select 'process parameters' and 'guide words' in sequence, to ensure a comprehensive examination of all parts of the system.

The usual Guide Words include; No, More, and Less but there are a number of others commonly used. However, not all Guide Words will be applicable to each parameter.

### 3.3.7.5 Causes

As each process parameter (e.g. Flow, Pressure, Temperature etc.) and guide word (e.g. No/None, More, Less etc.) is identified and applied, the combination of parameters and guide words in sequence will identify all the deviations<sup>4</sup> (e.g. no flow, more pressure, etc.).

The HAZOP Team should identify all the potential causes against a specific deviation within the node; causes must be limited to the confines of the Node under consideration. Frequently, there will be more than one cause, for example 'No Flow' would in all probability be caused by a blockage, but there may be several different causes of that blockage. If the generic cause alone is covered there is real possibility that some, methods of prevention and operational issues may be missed. To minimise that possibility, all the available causes of the deviation should be identified and analysed by the team.

<sup>1</sup> Refer to Appendix 4: Conditioning Factor Types.

<sup>2</sup> Refer to Appendix 3: Other Consideration Factors.

<sup>3</sup> Refer to Appendix 1: Parameters & Guide Words.

<sup>4</sup> Refer to Appendix 2: Deviations & Potential Causes.

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

Causes must be credible; a non-credible event would be one which is, possible but highly improbable. Causes should also not involve double jeopardy events, i.e. they should not involve two or more unrelated events or conditions that would have to occur concurrently in order to initiate the sequence of events that leads to adverse consequences. If non-credible causes are considered by the team this may lead to expensive and unnecessary modifications to the facility being implemented.

Causes shall be defined to include the 'why', the 'how', and where relevant, the 'when'. All causes must be verified to have valid initiating events, i.e. they need to have a unique cause-consequence relationship. Guidelines for identifying causes;

- Team brainstorm
- Address all types of causes e.g. human error, equipment failure, or external events
- Record all causes first prior to risk ranking
- Group equipment failures as appropriate
- Be specific i.e. blockage does not define method of prevention and operational issue, so blockage due to debris is more specific.
- Verify all causes are valid initiating events.

The three major categories of causes, in decreasing probability are: a) human error, b) equipment failure, c) external events. Team should consider the levels of definition of the causes to ensure adequate detail is provided. There are 3 levels of definition to be considered during HAZOP they are:

- Level 1 (immediate cause) e.g. Pump fails off.
- Level 2 (basic cause) e.g. Pump fails off due to Mechanical Failure.
- Level 3 (Enabling event or conditioning factor) e.g. Pump fails off due to mechanical failure due to lack of PM.

The Level 2 (basic cause) is to be regarded as a minimum and approach Level 3 wherever it is appropriate and possible.

### **3.3.7.6 Consequences**

The potential consequences related to each deviation should be discussed and assessed, within the limits of the information available and the expertise of the HAZOP Team.

Potential Consequences should assume the worst credible outcomes related to each risk category e.g. Personnel, Environment, PD-LOR (Property Damage and Loss of Revenue), and External impacts, without allowance for any existing safeguards.

The consequence outcome should not be limited to the confines of the Node under consideration. There may be several consequences involving escalation to other items of equipment, all such consequences should be evaluated using the Risk Matrix.

Consequence Rule Sets for Major Hydrocarbon Loss of Containment Scenario.

For all manned areas, the default consequence level for PAF caused by a Hydrocarbon Loss of Containment from a 100mm dia. orifice or greater shall be taken as 2 or 3 PII as a minimum.

### **3.3.7.7 Intrinsic Controls**

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

In QatarEnergy/ Idd El Shargi there are two types of intrinsic controls that may be considered, they are: CME (Critical Mitigation Element) and KME (Key Mitigation Element). Intrinsic controls (CMEs/KMEs) are those controls that may be assumed are or will be in place, e.g. provision of, ESD, fire and gas systems; PTW; design compliance with codes and Standards, operating Procedures etc.

Both the Intrinsic Controls (e.g. High Pressure Trip or Pressure Relief on a Vessel for overpressure protection), in place against each of the Hazard Scenarios and the Consequences (e.g. Loss of Containment (LOC), should be assessed to determine if the controls are robust and provide the necessary adequate barriers.

The HAZOP Team assesses the net Probability (likelihood), of the Hazard Event Scenario when factoring in the probability reduction provided by the associated Intrinsic Controls (CME's & KME's provided by the design and the operating Procedures). Note: if an Intrinsic Control is not included in the tabled key HAZOP engineering Documents/ drawings or is not physically in operation - then it shall not be regarded as an Intrinsic Control but may be considered as an Additional Control.

Use of Administrative type / Human intervention controls (KMEs), are generally not considered 'robust' controls and careful consideration must be undertaken if they are utilized to achieve Risk reduction from the C, D, E levels.

The intrinsic control types (CME or KME), are categorised according to their specific function. (i.e. CME-Eliminate, CME-Prevent, CME-Detect, CME- Mitigate, CME- Respond and KME-Eliminate, KME-Prevent, KME-Detect, KME-Mitigate, KME-Respond, KME-Recover); Refer to Fig 4.1 below.

| Intrinsic Controls: CMEs / KMEs | CAT | CME Name                   | HES Crit. Eqpm? | Initial Risk                |   |    |         |
|---------------------------------|-----|----------------------------|-----------------|-----------------------------|---|----|---------|
|                                 |     |                            |                 | C                           | P | RR | Comment |
|                                 |     |                            |                 |                             |   |    |         |
|                                 |     | Intrinsic Control Category |                 | Description                 |   |    |         |
|                                 |     | CME - Eliminate            |                 | Critical Mitigating Element |   |    |         |
|                                 |     | CME - Prevent              |                 | Critical Mitigating Element |   |    |         |
|                                 |     | CME - Detect               |                 | Critical Mitigating Element |   |    |         |
|                                 |     | CME - Mitigate             |                 | Critical Mitigating Element |   |    |         |
|                                 |     | CME - Respond              |                 | Critical Mitigating Element |   |    |         |
|                                 |     | KME - Eliminate            |                 | Key Mitigating Element      |   |    |         |
|                                 |     | KME - Prevent              |                 | Key Mitigating Element      |   |    |         |
|                                 |     | KME - Detect               |                 | Key Mitigating Element      |   |    |         |
|                                 |     | KME - Mitigate             |                 | Key Mitigating Element      |   |    |         |
|                                 |     | KME - Respond              |                 | Key Mitigating Element      |   |    |         |
|                                 |     | KME - Recover              |                 | Key Mitigating Element      |   |    |         |

FIGURE 4.1. INTRINSIC CONTROLS (CME/KME) CATEGORIES

Each CME must be independent and auditable. If, however, the HAZOP Team disagrees with the estimated probability result and considers the CME method unsuitable for the event scenario, then recourse to the other two methods of assigning probability (i.e., Industry / QRA approach), should be made. However, in such a case, the HAZOP Team should clearly define the basis for their assumptions and reasoning for the assigned probability level determined and concurrence sought by QatarEnergy/ Idd El Shargi Technical HSE. The discussion will be recorded in summary in the HAZOP Revalidation worksheets.

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

An AC: CME – Respond, may be credited for muster, escape and evacuation for offshore operations. This CME can be assigned only if personnel working in the plant area have full knowledge of escape routes, carry personal gas detectors, and have immediate access to BA sets and all necessary KME are in place, such as Procedures, training, testing and drills.

### **3.3.7.8 Initial Risk**

The Initial Risk should be determined, by means of the Team assigned Consequence and Probability levels, based on the identified, robust, Intrinsic Controls which determine the position on the QatarEnergy/ Idd El Shargi HSE Risk Matrix (i.e. the intersection point of the C/P). Note that when determining risk, the Consequence always take precedence over Probability.

### **3.3.7.9 Potentially Elevated Consequences (PEC)**

Once the initial risk has been determined, the team should identify and record the appropriate PEC Code <sup>5</sup> (PEC-1, PEC-2, N/A), some event scenarios have much higher consequences than the threshold value for a level 5 consequence. The risk evaluation team shall include PEC Codes where a consequence level of 5 is assigned for 'Personnel' and 'Property and Revenue' consequence categories as shown in Figure 3.2. If a PEC does not apply when a consequence level of 5 is determined the PEC Code "PEC N/A" shall be selected.

<sup>5</sup> Refer to: QP-ISND-RM-HSE-301 Risk Evaluation & Response Procedure

|                                                                                   |                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b> <span style="float: right;"><b>REV. 01</b></span> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

|           |                                                                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PEC Level | For 'Personnel' and 'Property and Revenue' with a Level 5 Consequence, one of the following PEC Codes shall be assigned. Note that the Hazard Scenario should not be modified to assign a PEC Code. |
| PEC 1     | At this Event Scenario's assigned Probability, the potential consequence exceeds 14 Permanent Incapacitating Injuries.                                                                              |
| PEC 2     | At this Event Scenario's assigned Probability, the potential consequence exceeds \$1 billion in combined PD/LOR.                                                                                    |
| PEC N/A   | PEC does not apply to this Event Scenario.                                                                                                                                                          |

FIGURE 4.2. POTENTIALLY ELEVATED CONSEQUENCE CODES

### 3.3.7.10 HSE Critical

In the event that a Consequence Level is determined to be a 4 or 5 then the equipment shall be considered as HSE Critical Equipment and identified as such by input of a 'Y' in the appropriate HSE Critical Equipment column of the HAZOP Worksheet<sup>6</sup> The related CMEs and KMEs will also be considered to be HSE Critical.

### 3.3.7.11 Additional Controls / Recommendations

The HAZOP Team have to consider what Additional Controls and Recommendations could be applied to reduce the Initial Risk to ALARP, and record any Recommendations for improvement which have been identified by the team. The identified Additional Controls (CME/KME), their category (i.e. CME - Eliminate, CME - Prevent, CME - Detect, CME - Mitigate, CME - Respond and KME - Eliminate, KME - Prevent, KME - Detect, KME - Mitigate, KME - Respond, KME - Recover) and the associated CME name, will be recorded together with the identified responsible action party.

| Additional Controls / Recommendations | CAT | CME Name                                   | Comment                     | Responsibility |
|---------------------------------------|-----|--------------------------------------------|-----------------------------|----------------|
|                                       |     |                                            |                             |                |
|                                       |     | Additional Control/Recommendation Category | Description                 |                |
|                                       |     | AC: CME - Eliminate                        | Critical Mitigating Element |                |
|                                       |     | AC: CME - Prevent                          | Critical Mitigating Element |                |
|                                       |     | AC: CME - Detect                           | Critical Mitigating Element |                |
|                                       |     | AC: CME - Mitigate                         | Critical Mitigating Element |                |
|                                       |     | AC: CME - Respond                          | Critical Mitigating Element |                |
|                                       |     | AC: KME - Eliminate                        | Key Mitigating Element      |                |
|                                       |     | AC: KME - Prevent                          | Key Mitigating Element      |                |
|                                       |     | AC: KME - Detect                           | Key Mitigating Element      |                |
|                                       |     | AC: KME - Mitigate                         | Key Mitigating Element      |                |
|                                       |     | AC: KME - Respond                          | Key Mitigating Element      |                |
|                                       |     | AC: KME - Recover                          | Key Mitigating Element      |                |
|                                       |     | Rec.                                       | Recommendation              |                |

FIGURE 4.3. ADDITIONAL CONTROLS AND RECOMMENDATIONS (AC/REC) CATEGORIES

<sup>6</sup> Refer to Appendix 5: Sample HAZOP Worksheets

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

Additional controls and recommendations will be SMART (Specific, Measurable, Assigned, Realistic and Time-bound). Additional controls and recommendations shall be written to be standalone (i.e. understandable without reference to the HAZOP Revalidation worksheets).

### **3.3.7.12 Residual Risk**

The 'Residual Risk', should be determined by the team by means of the modified probability or consequence levels based on the identified Additional Controls, which determine the potentially changed Risk position in the QatarEnergy/ Idd El Shargi Risk Matrix. In the event that no Additional Control or recommendation is identified then the Residual Risk shall be recorded on the worksheet at the same value as the Initial Risk.

Identified Recommendations for improvement should be documented within the report, and categorised as such in relevant CAT column of the HAZOP worksheet, but they will not provide any CME credit with respect to Residual Risk.

### **3.3.7.13 HAZOP Node Assessment Flowchart**

The flowchart in Figure 4.4 below shows the assessment process to be followed.



## HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE

DOC NO: IS-HSE-GDL-318

REV. 01

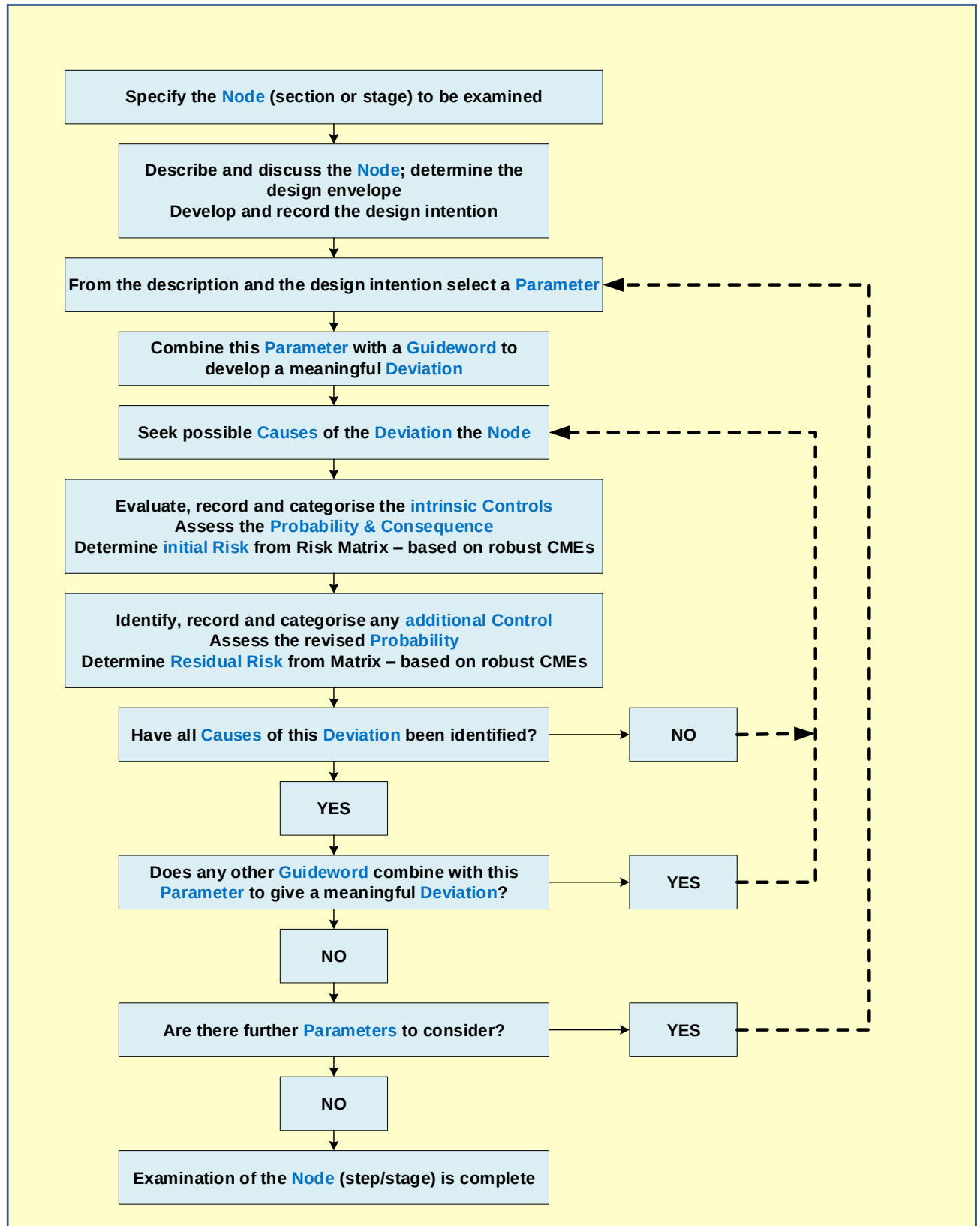


FIGURE 4.4. HAZOP NODE ASSESSMENT FLOWCHART

|                                                                                   |                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b> <span style="float: right;"><b>REV. 01</b></span> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.3.8 Reporting

Reporting should be, 'in full', rather than, 'by exception' e.g. The impact of low level in a vessel may be assessed with no consequence identified by the team, that evaluation will be recorded in the report to clearly identify for the records that the potential hazard was identified and assessed with no apparent issues identified. Individual Tags Numbers should be recorded as they are provided on the PEFD.

During the HAZOP, the Scribe will record within PHA-Pro all potential hazards, their Causes, Intrinsic Control (CMEs/KMEs), Consequence, Probability, Initial Risk, Additional Controls / Recommended Actions, Comments, modified Consequence, Probability and Residual Risk.

The following should be reported, to the QatarEnergy/ Idd El Shargi Technical HSE team for further evaluation.

- Risks at 'C w/PEC', 'D' & 'E' Level.
  - a. For Initial Risk levels of 'D' and 'E', where the team cannot identify any Additional Controls to reduce the Risk. (Note: A 'Recommendation' will be made in the HAZOP report to require further, more detailed review, with the Residual Risk level reported as the Initial Risk. This ensures the Risk level will be highlighted in the HAZOP Review / Summary report).
  - b. Where the residual Consequence level is 5, for any category of Risk.
- PEC Code 1 and PEC Code 2.

Potentially Elevated Consequence (PEC) Codes: Some event scenarios have much higher consequences than the threshold value for a level 5 consequence. The Review team should include PEC Codes where a consequence level of 5 is assigned and when:

- a. PEC 1 – Potential consequence to Personnel exceeds 14 Permanent Incapacitating Injuries.
- b. PEC 2 – Potential consequence in combined Property Damage (PD) and Loss of Revenue (LOR) exceeds \$1 billion. If a PEC code does not apply to the Event Scenario when a consequence level of 5 is determined the PEC Code, 'PEC N/A' should be applied.

For smaller HAZOPs, no larger than a few nodes, the 'HAZOP Summary Report' may be completed with the HAZOP Review Report (review) appended.

For larger HAZOPs, including plant revalidation HAZOPS, the Review report and close- out reports are to be issued, as well as 'HAZOP Summary Reports' on a regular basis.

The reports should be issued in line with the requirements of QP-ISND-RM-HSE-301 (Risk Evaluation and Response Procedure) and IS-HSE-GDL-303 (RM Administration Guideline).

A copy of all reports should be sent to the QatarEnergy/ Idd El Shargi RM Administrator and Process TA for record and distribution purposes.

The Standard report should contain:

- An Executive Summary
- A worksheet report (automatically generated by the PHA-Pro software.)

|                                                                                   |                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b> <span style="float: right;"><b>REV. 01</b></span> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

- A markup of P&IDs, signed by the PHR Facilitator/Facilitator

A completed review report can be used to demonstrate to interested parties that a process hazard analysis has been accomplished and all possible actions have been examined and or implemented to eliminate major hazards.

### 3.3.9 Records

A master copy of each P&ID will be signed and dated by the PHR Facilitator when all systems on that drawing have been examined and marked up. All the drawings, reference data, evaluations, record sheets, action sheets etc., used or developed during the course of the HAZOP should be reproduced and filed for future reference, by the PHR Administrator.

On completion of the HAZOP, the PHR Administrator will compile a report, describing the scope of the work and containing copies of the main Documents and drawings used. A typical report will contain an introduction, list of team members, dates of meetings, pertinent facts, actions summary and recommendations. Other principal Documents, including marked up P&IDs, Record sheets and important minutes of meetings and attendance record sheets will be presented as an Appendix. The final report is circulated and agreed to by all team members.

## 4.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

| Document Title                                                                                                                                            | Document Number | Internal/External Document (if any) | Document Referencing |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------|----------------------|
| Risk Evaluation & Response Procedure                                                                                                                      | IS-HSE-PRC-301  | Internal                            | Upwards              |
| Process Hazard Review Procedure                                                                                                                           | IS-HSE-PRC-308  | Internal                            | Upwards              |
| Critical Mitigation Element Methodology Guideline                                                                                                         | IS-HSE-GDL-309  | Internal                            | Sideways             |
| Petroleum and natural gas industries – Offshore production installations Guidelines on tools and techniques for hazard identification and risk assessment | ISO 17776: 2000 | External                            | N/A                  |

## 5.0 DEFINITIONS

TABLE 2: KEY TERMS DEFINED

|                                                                                   |                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b> <span style="float: right;"><b>REV. 01</b></span> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

| Term                       | Definition                                                                                                                                                                                                                                                                                                                                             |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROCESS HAZARD ANALYSIS    | An organized, thorough and systematic approach to identify and evaluate hazards associated with chemical processes and operations to enable their control.                                                                                                                                                                                             |
| HAZOP                      | The application of a systematic critical examination, to the process and engineering intentions, of a facility to assess the hazard and operability potential of mal-operation or malfunction of individual items of equipment and the consequential effects on the facility as a whole.                                                               |
| What-If-Analysis           | <p>What-If Study is a structured brainstorming method of determining what can go wrong and judging the likelihood and consequences.</p> <p>The answers to these questions form the basis for making judgments regarding the acceptability of those risks and determining a recommended course of action for those risks judged to be unacceptable.</p> |
| MANAGEMENT OF CHANGE (MOC) | MOC is a formal process that systematically manages permanent, temporary or emergency changes to the organizations, operations, process equipment, technologies, procedures or chemicals that could impact the original design, operations and maintenance criteria, and/or environmental performance.                                                 |

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

## APPENDIX

### APPENDIX A - PARAMETERS & GUIDE WORDS

Parameters:

- FLOW
- PRESSURE
- TEMPERATURE
- LEVEL
- VISCOSITY
- COMPOSITION
- CORROSION / EROSION
- RELIEF
- INSTRUMENTATION
- MAINTENANCE / MAINTAINABILITY
- UTILITIES
- DRAINAGE
- START-UP & SHUTDOWN
- OTHER (e.g. SAMPLING)

Typical Guide Words

| Guide Word              | Meaning                                                               | Comment                                                                                                                                                          |
|-------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NO, NOT or NONE         | The complete negation of the design intentions                        | No part of the intentions is achieved and nothing else happens.                                                                                                  |
| MORE (OF)<br>/LESS (OF) | Quantitative increase or decrease of any relevant physical parameters | These refer to quantities and relevant physical properties such as flow rates and temperatures as well as activities like 'HEAT' and 'REACTION'.                 |
| AS WELL AS              | A qualitative increase                                                | All the design and operating intentions are achieved together with some additional activity                                                                      |
| PART (OF)               | A qualitative decrease                                                | Only some of the intentions are achieved; some are not.                                                                                                          |
| REVERSE                 | The logical opposite of the intention                                 | This is mainly applicable to activities, for example reverse flow or chemical reaction. It can also be applied to substance, e.g. 'POISON' instead of 'ANTIDOTE' |

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

| Guide Word | Meaning               | Comment                                                                           |
|------------|-----------------------|-----------------------------------------------------------------------------------|
| OTHER THAN | Complete substitution | No part of the original intention is achieved. Something quite different happens. |

## APPENDIX B - DEVIATIONS & POTENTIAL CAUSES

A Deviation is a combination of Guide Word and Parameter

| Deviation        | Potential Causes                                                                                                                                                                                                                                                                |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No Flow          | Wrong routing, blockage (gas locking/foreign body), incorrect slip plate, incorrectly fitted/oriented N.R.V., ruptured pipe, large leak, equipment failure (C.V., isolation valve, pump, etc.), suction vessel empty, incorrect pressure differential, isolation in error, etc. |
| More Flow        | Increased pumping capacity, increased suction pressure, reduced delivery head, greater fluid density, exchanger tube leaks, restriction orifice plates deleted, cross connection of systems, control faults, etc.                                                               |
| Less Flow        | Line restriction, filter blockage, defective pumps, fouling of vessels, valves, restrictor or orifice plates, density or viscosity problems, incorrect specification of process fluid, etc.                                                                                     |
| More Pressure    | Surge problems, leakage from interconnected H.P. System, exchanger tube leak, gas breakthrough, isolation Procedures for relief valves defective, thermal overpressure, positive displacement pumps, failed open PCVs, etc.                                                     |
| Less Pressure    | Generation of vacuum condition, condensation, cavitation, restricted pump/compressor suction line, undetected leakage, vessel drainage, etc.                                                                                                                                    |
| More Temperature | Ambient conditions, chemical breakdown, fouled or failed exchanger tubes, fire situation, cooling water failure, defective control, fired heater control failure, reaction control failures, etc.                                                                               |
| Less Temperature | Ambient conditions, changes in viscosity, reducing pressure, fouled or failed exchanger tubes, loss of heating, etc.                                                                                                                                                            |
| High Level       | Outlet Isolated or Blocked, Inflow Greater Than Outflow, Control Failure' Faulty Level Measurement, Flooding, Pressure Surges, Corrosion, Sludge.                                                                                                                               |
| Low Level        | Inlet Flow Stops, Leak, Outflow Greater than Inflow, Control Failure, Faulty Level Measurement, Draining of Vessel.                                                                                                                                                             |

|                                                                                   |                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b> <span style="float: right;"><b>REV. 01</b></span> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

| Deviation                  | Potential Causes                                                                                                                                                                               |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| More/Less Viscosity        | Incorrect material specification, incorrect temperature, etc.                                                                                                                                  |
| Composition Change         | leaking isolation valves, leaking exchanger tubes, phase change, incorrect feedstock/specification, inadequate quality control, process control Procedures, etc.                               |
| Contamination / Impurities | Leaking exchanger tubes or isolation valves, incorrect operation of system-interconnected systems, effect of corrosion, wrong additives, ingress of air, shutdown and start-up conditions etc. |
| Corrosion / Erosion        | Cathodic protection arrangements, internal/external corrosion protection, engineering/material specifications, stress corrosion cracking, fluid velocities, etc.                               |

#### APPENDIX C - OTHER CONSIDERATION FACTORS

| Factor              | Consideration                                                                                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relief/vent         | Relief philosophy (process/fire), relief/valve discharge location, inlet and outlet piping, type of relief device, reliability, pollution implications, two phase flow, noise, etc.                                                                                                       |
| Instrumentation     | Location of instruments, response time, set points of alarms and trips and ability to change, time available for operator intervention, alarm and trip testing, trip/control amplifiers, panel arrangement and location, auto/manual facility and human error, fail safe philosophy, etc. |
| Sampling            | Sampling Procedure and operator safety, time for analysis result, calibration of automatic samplers, reliability/accuracy of representative sample, diagnosis of result, loss of sample flow, etc.                                                                                        |
| Maintenance         | Isolation philosophy, drainage, purging, flushing, cleaning, access, pressure testing, ISSOW, lifting and Manual handling, etc.                                                                                                                                                           |
| Utilities           | Failure of instrument air, steam, nitrogen, cooling water, electric power                                                                                                                                                                                                                 |
| Drainage            | Safe separation of Hazardous from non-hazardous drains, Isolation requirements for closed drains system, ISSOW, etc.                                                                                                                                                                      |
| Start-up & Shutdown | Normal shutdown, Emergency shutdown, isolation, purging, black start etc.                                                                                                                                                                                                                 |

#### APPENDIX D - CONDITIONING FACTOR TYPES

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

| Type                                                                                                                                                                             | Description                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| General Condition                                                                                                                                                                | General Condition of the node                                                                      |
| No Known Issues                                                                                                                                                                  | No Issues                                                                                          |
| Deferral: <ul style="list-style-type: none"> <li>• AI-MI</li> </ul> (Mechanical Integrity) <ul style="list-style-type: none"> <li>• AI- Mtce.</li> </ul> (Equipment Maintenance) | Inspection etc. (Fabric / Static equipment) Equipment Maintenance; Proof-test etc.                 |
| NRO (Non-Routine Operation)                                                                                                                                                      | Non-Routine Operation in the node or affecting the node function                                   |
| Organisational Change                                                                                                                                                            | Organisational Change                                                                              |
| Procedural Change                                                                                                                                                                | Procedural Change                                                                                  |
| RIK (Replacement-in-Kind)                                                                                                                                                        | Replacement in Kind                                                                                |
| Temporary Repair                                                                                                                                                                 | Wraps; Clamps; other- Technical Reviews                                                            |
| LTI (Long-term Isolation)                                                                                                                                                        | Long Term Inhibits- ESD, F&G, RV; Firewater                                                        |
| Related MOC's / MOC Requests                                                                                                                                                     | Related MOC's - planned or in-hand that may impact the review                                      |
| Process / Maintenance Overrides                                                                                                                                                  | Planned short-term overrides                                                                       |
| Historical Incidents                                                                                                                                                             | What? When? How? Why? Lessons Learned / Corrective actions?                                        |
| SIL assessed                                                                                                                                                                     | Have all the SIF (Safety Instrumented Functions) loops been SIL assessed and verified in the node? |
| No FTI                                                                                                                                                                           | Is all key Facilities Technical Information available and as-built for the node?                   |
| PHA-Actions Outstanding                                                                                                                                                          | PHA Actions which have not yet been implemented: FEC; OPS; Especially JSA-transient works          |
| Punch List Outstanding                                                                                                                                                           | Construction e.g. Temporary (wrong-specification material) bolts fitted; Cat. A; Cat. B            |

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><br><b>DOC NO: IS-HSE-GDL-318</b><br><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

| Type                                                                                        | Description                                                                                                                                                       |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Procedural Issue                                                                            | Procedure (NI; NC). Operations - chronic non-conformance or Procedure that does not exist or needs improvement                                                    |
| CAR (Corrective Action Request)                                                             | Audit; Incident - Various depts. All these - once accepted, if not actioned, constitute a subtle degradation of MOC                                               |
| No FTI                                                                                      | e.g. Weather; Stress; Competencies                                                                                                                                |
| Deficiency: <ul style="list-style-type: none"> <li>• AI-MI</li> <li>• AI- Maint.</li> </ul> | < MAWT; Leak; Structural Damage<br>Equipment / System Performance; Poor condition;<br>Blocked RV / Impulse Lines;<br>Difficult to maintain; 'Ex' eqpmt. condition |
| Operating Phase                                                                             | Change from Normal Ops., e.g. Start-up, shutdown, Emergency shutdown; Black start                                                                                 |
| Coincident Works                                                                            | Simops; Triops                                                                                                                                                    |
| Defect Design                                                                               | BV or Instrument access; Escape Routes compromised; Dead-legs; Risers External to jacket                                                                          |
| Personnel in Area                                                                           | Personnel in Area                                                                                                                                                 |



## HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE

DOC NO: IS-HSE-GDL-318

REV. 01

## APPENDIX E - SAMPLE HAZOP WORKSHEETS

## WORKSHEET COMPLETION GUIDANCE

Node: 1.  
Deviation : 1.

Design Intent  
Material:  
Activity:  
Conditioning Factors  
Type: Wraps, Long Term Inhibits, Deferrals, etc.

Automatically populated from the 'Node' sheet

Ensure Clear definitions of Node limits  
Deviation, (based on the Parameter and Guide Word under consideration)  
e.g. Parameter (Level) and Guide Word (Low), Deviation; Low Level,

Automatically populated from the 'Deviation' sheet

Automatically populated from the 'Node' sheet

Automatically populated from 'Conditioning Factors' Sheet

If C>3 then the CME and supporting KME are to be categorised as 'HSE Critical'

In the event that no Additional Controls are identified then the Residual Risk (RR) shall be recorded at the same value as the Initial Risk (IR)

| Cause                                                       | Hazard Scenario & Consequence                                                               | CAT                                                                  | Comments                                                                                                                   | Intrinsic Controls : CMEs / KMEs                        | CAT                                                                        | CME Name                                             | HSE Crit Eqmpt?                                                                 | Initial Risk |   |   |   | PEC Code                                                                                                            | Additional Controls / Recommendations                                            | CAT                                                                                 | CME Name                                                                   | Comment                                               | Responsibility                                                                 | HSE Crit. Eqmpt?                                      | Residual Risk |   |   |         |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------|--------------|---|---|---|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------|---------------|---|---|---------|
|                                                             |                                                                                             |                                                                      |                                                                                                                            |                                                         |                                                                            |                                                      |                                                                                 | C            | P | I | R |                                                                                                                     |                                                                                  |                                                                                     |                                                                            |                                                       |                                                                                |                                                       | C             | P | R | Comment |
| Provide Clear and concise description of Cause of Deviation | Provide clear and concise description of the Hazard Scenario and its potential consequences | Select Relevant Category from the drop down menu PAF/ER/PD/LOR\$/NOP | Document any significant comments which provide additional clarification of the Hazard Scenario and potential Consequences | Detail Intrinsic control relevant to the event scenario | Select Relevant Category from the drop down menu (Ref. Figure 7.7.1 above) | Select Relevant CME/KME Name from the drop down menu | If the consequence level is a 4 or 5 then a 'Y' shall be recorded in the column |              |   |   |   | Provide as necessary any comment clarifying why a CME may not be credited or Why the CME approach has not been used | Apply appropriate 'PEC Code' i.e. PEC1 PEC2 or PEC N/A (Ref. Figure 7.7.2 above) | Identify any robust additional controls or recommendations that the team identified | Select Relevant Category from the drop down menu (Ref. Figure 7.7.3 above) | Select Relevant CME/ KME Name from the drop down menu | Add comment where necessary to support or justify implementation of A/C or Rec | Identify responsible party i.e.FETL/FOT L/HSE Manager |               |   |   |         |
|                                                             |                                                                                             | PAF/ER/PD/LOR\$/NOP                                                  | Complete for each relevant CAT                                                                                             | Complete for each relevant CAT                          |                                                                            |                                                      |                                                                                 |              |   |   |   |                                                                                                                     |                                                                                  |                                                                                     |                                                                            |                                                       |                                                                                |                                                       |               |   |   |         |
|                                                             |                                                                                             | PAF/ER/PD/LOR\$/NOP                                                  | Complete for each relevant CAT                                                                                             | Complete for each relevant CAT                          |                                                                            |                                                      |                                                                                 |              |   |   |   |                                                                                                                     |                                                                                  |                                                                                     |                                                                            |                                                       |                                                                                |                                                       |               |   |   |         |
|                                                             |                                                                                             | PAF/ER/PD/LOR\$/NOP                                                  | Complete for each relevant CAT                                                                                             | Complete for each relevant CAT                          |                                                                            |                                                      |                                                                                 |              |   |   |   |                                                                                                                     |                                                                                  |                                                                                     |                                                                            |                                                       |                                                                                |                                                       |               |   |   |         |



## HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE

DOC NO: IS-HSE-GDL-318

REV. 01

## Example of Correct Worksheet Completion

Node: 1. Pipeline Gas KO Drum to Fuel Gas System  
 Deviation: 1. Low Level, Parameter (Level) and Guide Word (Low)  
 Design Intent  
 Material: Hydrocarbon Gas  
 Activity:

Conditioning Factors:  
 Type: No Known Issues

Provide clear and concise description of, Hazard Scenario including specific identifiers (relevant equipment Tag Nos.) and the potential Consequence related to the relevant Consequence Category

The comments provided document any significant information that provides additional clarification of the Hazard Scenario and potential Consequences.  
 For PAF issues the potential consequence should identify the related PII and not the Risk Matrix consequence level

Using the CME approach for Initial Risk each CME will reduce the Probability by one level. If in the below example there were three CMEs then the IR would reduce from 4, 3 IR D to 4, 2 IR C

In the event that no Additional Control or Recommendation is identified then the Residual Risk data should reflect the Initial Risk data

| Cause                                                                                                                        | Hazard Scenario and Consequence                                                                                                                                                                            | CAT                                                   | Comments                                                                                                                           | Intrinsic Controls: CMEs / KMEs                                                                                          | CAT                                                                        | CME Name                                             | HSE Crit Eqmpt?                                                                 | Initial Risk |   |   |   | PEC Code | Additional Controls / Recommendations                   | CAT                                                                        | CME Name                                             | Comment                                                                        | Responsibility                                        | HSE Crit. Eqmpt? | Residual Risk |   |   |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------|--------------|---|---|---|----------|---------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------|------------------|---------------|---|---|
|                                                                                                                              |                                                                                                                                                                                                            |                                                       |                                                                                                                                    |                                                                                                                          |                                                                            |                                                      |                                                                                 | C            | P | I | R |          |                                                         |                                                                            |                                                      |                                                                                |                                                       |                  | C             | P | R |
| One cause of Low Level may be:<br><br>1. Control loop failure of XV-XXXX-X (fail open)<br><br>(P&ID: PS1-X-XX-XXXX sht 001). | 1. Level will decrease in Gas KO Drum V-XXXX. Gas will blow by to Glycol regeneration system flash drum V-XXXX-1. Overpressure of V-XXXX-1 with potential for toxic gas and fire with injury to personnel. | Select appropriate Risk category PAF, EV, PD/LOR, NOP | Provide an appropriate Risk Cat PAF related Comment                                                                                | Detail Intrinsic control relevant to the event scenario. Two examples of Intrinsic Controls would be                     | Select Relevant Category from the drop down menu (Ref. Figure 7.7.1 above) | Select Relevant CME/KME Name from the drop down menu | If the consequence level is a 4 or 5 then a 'Y' shall be recorded in the column | 4            | 3 |   |   |          | PPE (Personal Gas Detector and SCBA) and escape routes. | Select Relevant Category from the drop down menu (Ref. Figure 7.7.3 above) | Select Relevant CME/KME Name from the drop down menu | Where necessary add comment to support or justify implementation of A/C or Rec | Identify responsible party i.e. FETL/FOTL/HSE Manager | 4                | 2             |   |   |
|                                                                                                                              |                                                                                                                                                                                                            | PAF                                                   | Team considered consequence of 2 or 3 permanent incapacitating injuries based on number of personnel present on the XX-XX platform | 1. PSV-XXXX-X on flash drum V-XXXX-1 designed for gas blow-by case.<br><br>2. Fire and gas detection and ESD on platform | CME Mitigate                                                               | PR-4                                                 | Y                                                                               |              |   |   |   | PEC N/A  |                                                         | CME Mitigate                                                               | TC-5                                                 | -                                                                              | FOTL                                                  |                  |               |   |   |
|                                                                                                                              |                                                                                                                                                                                                            | Select next appropriate Risk category EV, PD/LOR, NOP |                                                                                                                                    |                                                                                                                          | CME Mitigate                                                               | PR-6                                                 |                                                                                 |              |   |   |   |          |                                                         |                                                                            |                                                      |                                                                                |                                                       |                  |               |   |   |

|                                                                                   |                                                                                                             |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|  | <b>HAZARD &amp; OPERABILITY (HAZOP) REVIEW GUIDELINE</b><br><b>DOC NO: IS-HSE-GDL-318</b><br><b>REV. 01</b> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|

**APPENDIX F - REVISION HISTORY LOG**

TABLE 3: REVISION HISTORY LOG

Revision Number: 01

Date: 06/12/2023

| Item Revised: | Revision Description                                                            | Page No. |
|---------------|---------------------------------------------------------------------------------|----------|
| All           | Document simplified as part of QMR initiative. No change to technical contents. | All      |
| Remarks:      |                                                                                 |          |

**APPENDIX G - LIST OF ABBREVIATIONS**

TABLE 4: LIST OF ABBREVIATIONS

| Abbreviation | Description                        |
|--------------|------------------------------------|
| AI           | Asset Integrity                    |
| FTI          | Facilities Technical Information   |
| HAZOP        | Hazard & Operability Review        |
| MOC          | Management of Change               |
| PD/LOR       | Property Damage/ Loss of Revenue   |
| PEC          | Potentially Elevated Consequences  |
| PFD          | Process Flow Diagram               |
| P&ID         | Piping and Instrumentation Diagram |
| PHA          | Process Hazard Analysis            |
| PHR          | Process Hazard Review              |
| QRA          | Quantitative Risk Assessment       |
| SIL          | Safety Integrity Level             |